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COURSE 1.2 / COMPANION WORKBOOK

# Configure DNS and time

## Zones, records, the domain time hierarchy & troubleshooting

Build dependable internal name resolution and time on DC01 and DC02. Configure forward and reverse zones, manage service records, make the PDC emulator the domain's time source, and prove both in the air-gapped training network.

MATERIALS EDITION 1.2 / COURSE 1.2

Updated 8 October 2026 · sbd-dns-configuration

NAME _____ / DATE _____

# Table of contents

Read each lesson alongside the interactive guide and the video. Use the record tables to write down what you actually observe — the workbook is your evidence, not the guide.

Before you start

01 Map names to the network

02 Configure forward and reverse zones

03 Create and maintain service records

04 Prove resolution and directory discovery

05 Make SBD-DC01 the domain's time source

06 Find the fault and record the fix

Course review and sign-off

> **HOW TO USE IT** Each lesson page has the same shape: what you will learn, the build steps, the commands that verify them, and a record table. Fill the table from your own environment — never from the sample values.

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# Before you start

The addresses and names below continue the training example. Match every change to your own address plan. This network has no internet DNS or time dependency: names resolve on the domain controllers, and time comes from the PDC emulator, SBD-DC01.

## Prerequisites

- ☐ Complete the DC01 + DC02 identity build with both domain controllers healthy.
- ☐ Use an isolated Windows Server 2025 practice environment and an account delegated to manage DNS.
- ☐ Continue the existing sbd.test example: SBD-DC01 is 10.0.10.10 and SBD-DC02 is 10.0.10.11 on 10.0.10.0/24.
- ☐ The network's routing and VLAN segmentation are already in place, with a documented service-path register.

## What this course builds

- Plan authoritative internal DNS and client settings.
- Configure secure AD-integrated zones and maintain records.
- Test both resolvers, directory discovery, and client fallback.
- Make the PDC emulator the domain's time source and verify the time hierarchy end to end.

## Systems in scope

| RECORD | YOUR OBSERVATION |
|---|---|
| SBD-DC01 — name, address, OS build | |
| SBD-DC02 — name, address, OS build | |

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LESSON 01 / ABOUT 15 MIN

# Map names to the network

**WHY THIS MATTERS** A DNS plan explains which server owns a name and how a client finds that server. Begin with the working directory build, then extend it deliberately.

## In this lesson you will

- Explain which DNS servers a domain member must use in this lab and why.
- Inventory the existing sbd.test and _msdcs zones without creating duplicates.
- Document the forwarding and recursion decision for an air-gapped network.

## Follow the build

1. Record the domain, each subnet, both DNS addresses, and the owner of each service name. Keep host records and service aliases distinguishable.
2. On each domain controller, open DNS Manager and review the existing sbd.test and _msdcs zones. Promotion may already have created them; do not create duplicate zones.
3. Check the DNS role and static network settings. Domain members must use internal DNS servers that resolve the AD namespace; a public resolver is not an alternate for this environment.
4. Decide whether any separate internal namespace needs a conditional forwarder. With no recursive dependencies, document a no-recursion design; do not add internet forwarders or delete every root hint as a shortcut.

## Check your result

```
Get-WindowsFeature DNS
Get-DnsClientServerAddress -AddressFamily IPv4
Get-DnsServerZone -ComputerName SBD-DC01
```

**WHAT SUCCESS LOOKS LIKE** Both servers and the client point to the intended internal DNS service. Existing AD zones and the forwarding/recursion decision are recorded.

## Your record

Evidence to keep: Save the address/name worksheet and the initial zone inventory before changing anything.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |
| Evidence file names | |

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| RECORD | YOUR OBSERVATION |
| --- | --- |
| Open items or findings | |

## Think it through

Which DNS servers should a domain member use in this lab?

> **REMEMBER** Domain members point only at the internal DNS servers that resolve sbd.test. The existing AD zones are inventoried, not recreated. The forwarding and recursion decision is written down before anything changes.

Microsoft · Install and configure DNS Server

LESSON 02 / ABOUT 20 MIN

# Configure forward and reverse zones

**WHY THIS MATTERS** Forward lookup answers where a name lives. Reverse lookup answers which name is assigned to an address. AD integration lets the directory replicate zone data between the selected DNS servers.

## In this lesson you will

- Confirm sbd.test is AD-integrated with the intended replication scope.
- Create a secure, AD-integrated reverse zone for 10.0.10.0/24.
- Explain how zone data moves between the two DNS hosts.

## Follow the build

1. In DNS Manager, inspect sbd.test Properties. Confirm that it is AD-integrated and choose the domain or forest replication scope that matches the DNS servers intended to host it.
2. Use secure dynamic updates for the AD-integrated zone. Review delegation and permissions before broadening who can change records.
3. Under Reverse Lookup Zones, create an IPv4 primary AD-integrated zone for network ID 10.0.10 if an equivalent zone does not already exist. Select the approved replication scope and secure updates.
4. Review zone-transfer settings. AD replication between these DCs does not require enabling transfers to arbitrary DNS servers; restrict any separately required secondary-server transfers explicitly.

## Check your result

```
Get-DnsServerZone -ComputerName SBD-DC01 |
  Select-Object ZoneName, ZoneType, IsDsIntegrated, DynamicUpdate, ReplicationScope
Get-DnsServerZone -ComputerName SBD-DC02
```

**WHAT SUCCESS LOOKS LIKE** The intended forward and reverse zones appear on both DNS hosts after replication. The reverse zone for this /24 is 10.0.10.in-addr.arpa.

## Your record

Evidence to keep: Capture the zone type, update policy, replication scope, and transfer policy for each zone.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |
| Evidence file names | |

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| RECORD | YOUR OBSERVATION |
| --- | --- |
| Open items or findings | |

## Think it through

How does an AD-integrated zone reach the other DNS-enabled domain controller in its replication scope?

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> **REMEMBER** Forward and reverse zones are AD-integrated with a deliberate replication scope. Dynamic updates are secure-only. Zone transfers stay restricted; directory replication moves the data.

Microsoft · Manage DNS zones

LESSON 03 / ABOUT 20 MIN

# Create and maintain service records

**WHY THIS MATTERS** Build a small, intentional record set. A host record identifies a machine; an alias provides a stable service name without making that alias a separate server.

## In this lesson you will

- Distinguish host records (A and PTR) from service aliases (CNAME).
- Create a host record with its PTR and a service alias that match the plan.
- Maintain records with recorded ownership, lifetimes, and reasons.

## Follow the build

1. Review the DC A and PTR records first. Correct mismatches through the owning host and DNS administration process rather than blindly adding another record with the same name.
2. Choose one approved service host from your address plan. In sbd.test, use New Host (A or AAAA), enter its actual name and address, and select the associated PTR option when the reverse zone is ready.
3. For a service alias, use New Alias (CNAME) and select the intended fully qualified host. For the later PKI course, reserve pki.sbd.test for the approved internal publication host; do not point it at the offline root.
4. Set the record lifetime to the operational plan, record static ownership, and remove only records confirmed obsolete. An alias cannot coexist with other record types at the same name.

## Check your result

```
Get-DnsServerResourceRecord -ComputerName SBD-DC01 -ZoneName sbd.test
Resolve-DnsName SBD-DC01.sbd.test -Type A -Server 10.0.10.10
Resolve-DnsName 10.0.10.10 -Type PTR -Server 10.0.10.10
```

**WHAT SUCCESS LOOKS LIKE** Forward and reverse responses agree with the address plan. Each alias reaches the intended host; creating a DNS name alone does not configure its service.

## Your record

Evidence to keep: Keep a record table with owner, type, target, time to live, and reason for the change.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |
| Evidence file names | |

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| RECORD | YOUR OBSERVATION |
| --- | --- |
| Open items or findings | |

## Think it through

What does creating pki.sbd.test as an alias accomplish?

> **REMEMBER** Host records identify machines; aliases provide stable service names. pki.sbd.test is reserved for the internal publication host, never the offline root. Every record carries an owner, a lifetime, and a reason.

Microsoft · Manage DNS resource records

LESSON 04 / ABOUT 20 MIN

# Prove resolution and directory discovery

**WHY THIS MATTERS** Test each server explicitly before testing client fallback. A successful cached answer does not prove that both DNS servers hold the right records.

## In this lesson you will

- Query each DNS server explicitly instead of trusting cached answers.
- Verify the LDAP service records that let clients discover domain controllers.
- Run a controlled failover test and record the result.

## Follow the build

1. From a domain client, query each DNS server for the same host and reverse record. Record the server queried and the returned data.
2. Query the AD LDAP service records below against both servers. These records let a client discover domain controllers; an ordinary host lookup does not prove that discovery works.
3. Run the DNS diagnostics and directory replication checks on a domain controller. Investigate failures in context, including the intentionally isolated network design.
4. During an approved lab test, make one DNS server unavailable to the test client and verify resolution through the other. Restore connectivity and repeat for the opposite server. Do not interrupt shared production services for this exercise.

## Check your result

```
Resolve-DnsName SBD-DC01.sbd.test -Server 10.0.10.10
Resolve-DnsName SBD-DC01.sbd.test -Server 10.0.10.11
Resolve-DnsName _ldap._tcp.dc._msdcs.sbd.test -Type SRV -Server 10.0.10.10
Resolve-DnsName _ldap._tcp.dc._msdcs.sbd.test -Type SRV -Server 10.0.10.11
dcdiag /test:dns /v
repadmin /replsummary
```

**WHAT SUCCESS LOOKS LIKE** Both DNS servers return the expected host and AD service records, replication is healthy, and the client still resolves internal names when either resolver is unavailable.

## Your record

Evidence to keep: Keep paired lookup results, diagnostic output, and the failover test record with start, restore, and final verification times.

| RECORD | YOUR OBSERVATION |
| --- | --- |
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |

| RECORD | YOUR OBSERVATION |
| --- | --- |
| Evidence file names | |
| Open items or findings | |

## Think it through

Why query each DNS server directly?

**REMEMBER** Both resolvers answer the host and SRV queries correctly. dcdiag and repadmin confirm DNS and replication health. The failover test is documented with times, not just a passing screenshot.

Microsoft · Resolve-DnsName

Microsoft · Dcdiag

LESSON 05 / ABOUT 30 MIN

# Make SBD-DC01 the domain's time source

> **WHY THIS MATTERS** Kerberos tolerates minutes of skew, not more, and an air-gapped network has no internet time to lean on. Make the PDC emulator, SBD-DC01, the domain's authoritative time source and let the domain hierarchy carry its time to everything else.

## In this lesson you will

- Configure SBD-DC01, the PDC emulator, as the domain's reliable time source.
- Keep every other system on the domain hierarchy, with nothing overriding it.
- Document how DC01's clock is set and checked, and the drift threshold the model calls for.

## Follow the build

1. Confirm which domain controller holds the PDC emulator role — SBD-DC01 in this sample. The PDC emulator of the forest root domain sits at the top of the Windows Time hierarchy, so it is the one system you configure; everything else follows it.
2. On SBD-DC01, mark Windows Time as a reliable time source with no external peer, so it serves time from its own clock. In an air-gapped network that clock is the reference by declaration: record how it is set and checked — for example, compared on a schedule against an approved reference and corrected by an administrator, with each correction recorded — and the drift threshold you will hold it to.
3. Turn off the hypervisor's time synchronization for both domain controller virtual machines, so the host clock cannot silently override the PDC emulator. SBD-DC02 syncs from the PDC emulator, every other domain member keeps the default domain-hierarchy sync from the domain controllers, and the domain-joined Linux servers in later courses point chrony or ntpd at both domain controllers.
4. Confirm the network's service-path register permits time — every member to both domain controllers on UDP 123 — and write the time design into the build record: course 1.3's hardening Group Policy objects (GPOs) must leave it intact, and if a Security Technical Implementation Guide (STIG) value disagrees with it, that course's deviation record is where the decision lives.
5. Verify the chain end to end: SBD-DC01 serving as the reliable source, SBD-DC02 and a member server sourcing from a domain controller, and offsets within your recorded threshold.

## Check your result

```
Get-ADDomain | Select-Object PDCEmulator   # expected: SBD-DC01.sbd.test
# On SBD-DC01 (the PDC emulator): serve time from its own clock as the reliable source
w32tm /config /reliable:yes /update
Restart-Service w32time
w32tm /query /source   # on SBD-DC01: Local CMOS Clock
# On SBD-DC02 and a member server:
w32tm /query /source   # expected: a domain controller
w32tm /query /status
w32tm /stripchart /computer:SBD-DC01.sbd.test /samples:3 /dataonly
```

> **WHAT SUCCESS LOOKS LIKE** SBD-DC01 reports itself as the reliable source (Local CMOS Clock), SBD-DC02 and the member server report a domain controller, host time synchronization is off for both domain controllers, and measured offsets sit inside the documented threshold.

## Your record

Evidence to keep: Record the time design decisions (how DC01's clock is set and checked, and the drift threshold), the source and status output from both controllers and the member, the stripchart offsets, the hypervisor setting for both domain controller VMs, and the register rows that permit time: every member to the domain controllers.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |
| Evidence file names | |
| Open items or findings | |

## Think it through

Which system is the domain's authoritative time source in this design?

**REMEMBER** SBD-DC01, the PDC emulator, is the declared source; how its clock is checked is written down. The hypervisor no longer overrides domain controller time; the hierarchy carries it everywhere. Offsets are measured and inside the threshold, not assumed.

Microsoft · Windows Time service

Microsoft · Windows Time service tools and settings

LESSON 06 / ABOUT 15 MIN

# Find the fault and record the fix

**WHY THIS MATTERS** Troubleshoot one layer at a time: client settings, server reachability, authoritative data, replication, and finally the application that uses the name.

## In this lesson you will

- Isolate a DNS failure one layer at a time.
- Distinguish a missing name (NXDOMAIN) from timeouts and service failures.
- Record the cause, the fix, and the rollback — not just a passing result.

## Follow the build

1. For a failing lookup, record the exact name, record type, client, resolver, timestamp, and error. Compare the fully qualified name with any short-name search behavior.
2. Query both servers directly. If they disagree, inspect zone scope and replication. If both are correct but the client is not, inspect client DNS settings and cached data.
3. Distinguish a missing name (NXDOMAIN) from a timeout or service failure. Fix the owning configuration first, then use Clear-DnsClientCache on the test client if stale cache remains relevant.
4. Repeat the original test and the redundancy checks. Record what changed and the rollback method. Review aging/scavenging separately against DHCP, static records, and backup practices before enabling automatic deletion.

## Check your result

```
Get-DnsClientServerAddress -AddressFamily IPv4
Get-DnsClientCache
Resolve-DnsName SBD-DC02.sbd.test -Server 10.0.10.10
Resolve-DnsName SBD-DC02.sbd.test -Server 10.0.10.11
```

**WHAT SUCCESS LOOKS LIKE** The original failure is reproducibly resolved, both resolvers agree, and the change record explains the cause rather than only showing a passing screenshot.

## Your record

Evidence to keep: Complete the DNS checklist and retain the before/after lookup results with the configuration owner.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |

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| RECORD | YOUR OBSERVATION |
| --- | --- |
| Evidence file names | |
| Open items or findings | |

## Think it through

Both DNS servers return the correct address, but one client keeps using the old address. What should you inspect next?

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> **REMEMBER** Troubleshoot client settings, reachability, authoritative data, replication, then the application. NXDOMAIN, timeouts, and service failures point at different owners. The change record explains the cause and the rollback.

Microsoft · Manage DNS resource records

Microsoft · Resolve-DnsName

# Course review and sign-off

Before moving on, confirm each lesson left evidence behind — and that what remains open is written down, not remembered.

## Lesson completion

| LESSON | EVIDENCE FILED | DATE |
|---|---|---|
| 01 Map names to the network | ☐ yes | |
| 02 Configure forward and reverse zones | ☐ yes | |
| 03 Create and maintain service records | ☐ yes | |
| 04 Prove resolution and directory discovery | ☐ yes | |
| 05 Make SBD-DC01 the domain's time source | ☐ yes | |
| 06 Find the fault and record the fix | ☐ yes | |

## Open items carried forward

## Next in the learning path

That completes Configure DNS and time. Course 1.3, Harden the baseline, builds the organizational units and hardened security policies that every later system joins.

> **KEEP IT HONEST** Lesson completion records learning activity; it is not a system authorization decision. Your observations, not the sample values, establish what happened in your environment.